

Department of Electrical and Computer Engineering
University of Wisconsin – Madison
ECE 552 Introduction to Computer Architecture
In-Class Exercise (11/19)

1. Consider the following MIPS assembly program consisting of three threads, running on a **three-core shared-memory multiprocessor**. Assume that all memory locations initially contain the value 0, and the processor cores are in-order.

```
# initially for all threads:
#      [$s1] = 0x1110, [$s2] = 0x2220, [$s3] = 0x3330

# thread A
A0:   addi $t8, $zero, 123
A1:   sw   $t8, 0($s2)           # (0x2220) ← 123
A2:   addi $t8, $zero, 1
A3:   sw   $t8, 0($s1)           # (0x1110) ← 1
...

# thread B
B0:   lw   $t3, 0($s2)           # [$t3] ← (0x2220)
B1:   addi $t2, $zero, -123      # [$t2] ← -123
B2:   lw   $t8, 0($s1)
B3:   beq  $t8, $zero, B6        # if (0x1110) ≠ 0 then...
B4:   lw   $t2, 0($s2)           # ...[$t2] ← (0x2220)
B5:   sw   $t8, 0($s3)           # ... (0x3330) ← (0x1110)
B6:   ...

# thread C
C0:   lw   $t4, 0($s1)           # [$t4] ← (0x1110)
C1:   lw   $t0, 0($s3)           # [$t0] ← (0x3330)
C2:   lw   $t1, 0($s1)           # [$t1] ← (0x1110)
...
```

- a) Assume that there are **no private data caches** on this multiprocessor. Which of the following register outcomes are **possible** after executing all these instructions (\$t0, \$t1 and \$t4 in thread C; \$t2 and \$t3 in thread B)? Select all that apply.

- i. [\$t0] = 0, [\$t1] = 0, [\$t2] = 123, [\$t3] = 0, [\$t4] = 0
- ii. [\$t0] = 1, [\$t1] = 1, [\$t2] = 123, [\$t3] = 0, [\$t4] = 0
- iii. [\$t0] = 1, [\$t1] = 0, [\$t2] = 123, [\$t3] = 0, [\$t4] = 0
- iv. [\$t0] = 0, [\$t1] = 0, [\$t2] = 0, [\$t3] = 0, [\$t4] = 0
- v. [\$t0] = 1, [\$t1] = 0, [\$t2] = 0, [\$t3] = 0, [\$t4] = 0

Answer:

i and ii

Since [\$t2] cannot be 0, and if [\$t0] is 1 then [\$t1] must also be 1.

- b) Assume each thread is running on a separate core, each with a **private L1 data cache**. The caches are fully-associative and can fit three blocks; each cache block is 4B in size. The caches use a **write-through, write-allocate** policy and are initially empty.

Cache coherence is not implemented. The program happens to perform its memory accesses in the following order: B0, C0, A1, A3, B2, B4, B5, C1, C2. Fill in the table below, specifying the **data contents** of all caches (and main memory) **after** each access is performed. If a block is not in the cache, leave the corresponding cell blank. Also, at the bottom of the table, specify if each access is a cache hit (H) or miss (M). The first three accesses have been filled in for you.

Memory Access Sequence (time→)										
	init	B0	C0	A1	A3	B2	B4	B5	C1	C2
Thread A Private Cache										
0x1110					1	1	1	1	1	1
0x2220				123	123	123	123	123	123	123
0x3330										
Thread B Private Cache										
0x1110						1	1	1	1	1
0x2220		0	0	0	0	0	0	0	0	0
0x3330								1	1	1
Thread C Private Cache										
0x1110			0	0	0	0	0	0	0	0
0x2220										
0x3330									1	1
Main Memory										
0x1110	0	0	0	0	1	1	1	1	1	1
0x2220	0	0	0	123	123	123	123	123	123	123
0x3330	0	0	0	0	0	0	0	1	1	1
H/M?		M	M	M	M	M	H	M	M	H

- c) What are the contents of registers \$t0, \$t1, \$t2, \$t3 and \$t4 after this program execution (Q1b) on the multiprocessor **without cache coherence**?

Answer:

[\$t0] = 1, [\$t1] = 0, [\$t2] = 0, [\$t3] = 0, [\$t4] = 0

- d) Assume the same multiprocessor as Q1b, with private L1 data caches. This time, **cache coherence is implemented**. Assume that the coherence protocol is **invalidation-based**: multiple caches can hold a copy of a block, but when one cache modifies it, all other caches that currently have this block immediately invalidate (evict) it.

Fill in the table below for the same sequence as in Q1b, specifying the **data contents** of all caches (and main memory) **after** each access is performed. If a block is not in the cache, leave the corresponding cell blank. Also, at the bottom of the table, specify if each access is a cache hit (H) or miss (M).

Memory Access Sequence (time→)										
	init	B0	C0	A1	A3	B2	B4	B5	C1	C2
Thread A Private Cache										
0x1110					1	1	1	1	1	1
0x2220				123	123	123	123	123	123	123
0x3330										
Thread B Private Cache										
0x1110						1	1	1	1	1
0x2220		0	0				123	123	123	123
0x3330								1	1	1
Thread C Private Cache										
0x1110			0	0						1
0x2220										
0x3330									1	1
Main Memory										
0x1110	0	0	0	0	1	1	1	1	1	1
0x2220	0	0	0	123	123	123	123	123	123	123
0x3330	0	0	0	0	0	0	0	1	1	1
H/M?		M	M	M	M	M	M	M	M	M

- e) What are the contents of registers \$t0, \$t1, \$t2, \$t3 and \$t4 after this program execution (Q1d) on the multiprocessor **with cache coherence**?

Answer:

[\$t0] = 1, [\$t1] = 1, [\$t2] = 123, [\$t3] = 0, [\$t4] = 0